

Employment

Graduate Research Assistant	NC State University	April 2025 – Present
- Deployed containerized genomic platform on GCP Cloud Run (Docker, Cloud Build CI) with JBrowse 2 genome browser; enabled 100+ researchers to visualize 500GB+ datasets by eliminating on-premise infrastructure. - Improved ML model accuracy by 12% via grid-search hyperparameter sweeps and sliding-window feature engineering on multivariate biological sensor time-series.		
Product Engineer Intern	Isomer AI	May 2025 – Present
- Shipped 10+ production AI features end-to-end: multi-model LLM pipelines (Claude, GPT-4o, Gemini), Temporal.io agentic workflows, and REST APIs; owned product from spec to deployment across 923 commits and 7 releases. - Owned PostgreSQL, Redis, Auth0 RBAC, Terraform, Docker, and Sentry stack; triaged and resolved production incidents by tracing failures across async microservices.		
Software Developer Intern	Mphasis (HP Company)	Jun 2024 – Aug 2024
- Improved document processing accuracy by 25% by building OCR + NLP classification pipelines that parsed unstructured health claim documents into structured fields for downstream automation. - Automated AWS and GCP releases via GitHub Actions CI/CD; configured Grafana dashboards for observability and led red-teaming to surface model reliability gaps pre-production.		
AI/ML Intern	Capgemini	Jun 2023 – Aug 2023
- Reduced ticket resolution time by 15% by training an NLP multi-class classifier that auto-routed support tickets to the correct team based on predicted category and urgency. - Cut ticket-processing latency by 20% by replacing synchronous DB polling with Kafka async ingestion feeding Spark micro-batch jobs across partitioned Hadoop HDFS.		

Education

Raleigh, NC	North Carolina State University	Aug 2024 – May 2026
- Master of Science in Computer Science - Courses: Artificial Intelligence, Operating Systems, Algorithms, Data Science, Object Oriented Development, Software Engineering.		
Mumbai, India	Mumbai University	Aug 2020 – May 2024
Bachelor of Technology in Computer Science; CGPA: 9.32/10. Courses: Data Warehousing, Big Data Infra, DS & Algorithms, Machine Learning, Blockchain.		

Projects

Expertiza	Ruby on Rails Open Source Contribution	2024–2025
- Refactored core controllers in production Ruby on Rails codebase (SOLID + DRY principles); extracted business logic into service layers, eliminating copy-pasted patterns; cut code redundancy by 35%. - Added 110+ RSpec tests covering happy path and edge cases; increased coverage by 25% on production platform with 5,000+ users and 300+ contributors.		
Glance AI	Privacy-First RAG for Healthcare	2025
- Designed per-tenant RAG pipeline for healthcare with strict vector isolation preventing cross-tenant data leakage and LLM hallucinations. - Implemented deterministic retrieval for regulated environments; eliminated context bleed between tenants for factual LLM outputs.		

Languages and Technologies

- Languages:** Ruby, Python, TypeScript, JavaScript, Java, SQL, Bash
- Backend & APIs:** Ruby on Rails, RSpec, PostgreSQL, REST APIs, Node.js, FastAPI, Express, API design, webhooks
- AI & Data:** LLMs (Claude, GPT-4o, Gemini), RAG, Temporal.io, Kafka, Elasticsearch, data pipelines
- Infrastructure:** AWS, GCP, Docker, Kubernetes, Terraform, GitHub Actions, Redis, Grafana

Additional Experience and Awards

- IEEE Published:** Co-authored paper on CV-Based Cybersecurity Threat Detection (Springer icSoftComp 2023); presented to 500+ attendees.
- National Hackathon Winner:** Won or ranked Top 10 in 6+ national hackathons spanning ML, systems, and developer tools; built 3 complete MVPs end-to-end within 48-hour sprints, each with a working demo.
- Lead at Google Developers Student Club:** Led 4+ tech events (hackathons, conferences) reaching 500+ students.
- Technical Writing:** Published on LLM model collapse, AI agent security, audio AI at kulkarniaditya.com/blog